

WHICH BIAS FLAGS CAN YOU TRUST? A RELIABILITY MAP FOR FRONTIER-MODEL NEWS-BIAS DETECTION

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ABSTRACT

News-bias products—bias checkers, balanced-news feeds, media-literacy tools—are increasingly built by asking a frontier model to flag biased spans, label an article’s lean, and explain itself. But which of those outputs can a builder trust? We evaluate two frontier target models and two cross-family judges on 200 expert-rated U.S. news articles and find a sharp split. *Trustworthy* primitives: the *volume* of flagged spans is a valid meter of an article’s partisan intensity (Spearman $\rho \approx 0.6$ vs. expert ratings), and cross-model *agreement* predicts correctness cleanly (two models agreeing on lean are 74% accurate vs. 22% when they disagree; four agreeing, 87%). *Untrustworthy* primitives: a single model’s lean label near the center-right, its “bias by omission” or “unsubstantiated claim” flags (chance-corrected agreement $\kappa \leq 0.14$), and the specific text it highlights (cross-model overlap 0.26) are model-specific opinion, not detected fact. We also document a directional failure mode shared by all four models—moderate-right articles are read as center or *left*, so a lean-labeling tool built on them systematically under-counts right-leaning content—and translate the map into deployable triage rules: an agreement-based router clears 43% of volume at 85% accuracy while catching 86% of errors. Findings are exploratory (not pre-registered) and inherit AllSides outlet labels as ground truth.

1 INTRODUCTION

A growing class of products puts a frontier language model in the role of a news-bias analyst: it highlights loaded spans, assigns a left/right lean, summarizes “both sides,” and narrates why. The implicit assumption is that these outputs are measurements. Many are not—they are one model’s opinion, and a different frontier model, shown the same article, would say something else. A builder needs to know, before shipping, *which* outputs reproduce and which do not, and how to triage the difference.

This paper supplies that reliability map. We do not ask whether the models are biased (a companion paper takes up the behavioral question); we ask which of a bias detector’s outputs a tool can present to a reader as reliable. The answer is a clean split between primitives anchored to explicit text—which reproduce across independent models—and primitives that require inference about what is absent, implied, or where an article sits on a scale—which do not. We close with the practical form of the map: routing rules, and one systematic, directional error a builder must correct for.

Contributions.

1. **A reliability map of bias-detection primitives.** Which outputs of a frontier-model bias detector reproduce across models (detection volume; cross-model agreement) and which are model-specific opinion (lean labels near center-right; omission and framing flags; specific highlights)—each with a number.
2. **A directional failure mode shared across vendors.** All four models read moderate-right content as center-or-left, so a lean-labeling tool built on them under-counts the right; we localize the hardest class, topics, and per-model detection habits.

3. **Deployable triage rules.** Agreement-based routing and a primitive-by-primitive display policy that a product can adopt directly, plus a $\sim \$150$ per-model-upgrade regression audit.

2 SETUP

We reuse the corpus, models, and rollouts of the companion study; we summarize here and defer the full protocol (including the reliability gates behind the judge instruments) to it. The corpus is $N=200$ contemporary U.S. news articles on a balanced 5×8 grid: five AllSides-anchored lean classes (Left, Lean Left, Center, Lean Right, Right; 40 each) $\times$ eight topics, models receiving article text only (no outlet or headline). Two frontier *target* models (Claude Sonnet 4.5, GPT-4.1) and two cross-family *judge* models (Claude Sonnet 4.6, GPT-5) perform three tasks: span-level bias detection with a shared bias-type schema (Spinde et al., 2021; Hamborg et al., 2019), summarization with represented-perspective counts, and five-class lean classification with confidence. All analyses below are exploratory and use expert AllSides labels as ground truth (Baly et al., 2020).

3 WHICH PRIMITIVES REPRODUCE

Detection volume is a valid intensity meter. The simplest possible signal—how many spans a model flags—tracks how loaded an article actually is: flagged-span count rises monotonically with the article’s expert-rated partisan intensity (Spearman $\rho=0.64$ for Sonnet, 0.58 for GPT-4.1 against |lean rating|, both $p<10^{-18}$; partisan articles draw $2.3\times$ the flags of Center articles). A tool can trust a flag *count* as a bias meter even where it cannot trust the individual flags.

Cross-model agreement predicts correctness. The strongest cheap reliability signal is two independent models agreeing. On five-class lean, when the two target models agree they are 74% accurate; when they disagree, only 22% ($n=199$ shared articles; they agree 61% of the time). Adding the two judges sharpens it into a ladder: when all four models agree (44% of articles) the modal label is 87% accurate, falling monotonically to 75% at three-of-four and 62% at two (Table 1). Agreement is a free, ground-truth-independent confidence signal.

Models agreeing on the label	Share of articles	Accuracy
4 of 4 (unanimous)	44%	0.87
≥ 3 of 4	74%	0.75
≥ 2 of 4	99%	0.62

Table 1: Multi-model agreement is a monotone accuracy signal. Four frontier models (two families, targets and judges); modal-label accuracy vs. expert AllSides labels.

Only concrete, word-anchored bias forms reproduce. Given both target models the same bias-type schema, we ask how often they agree a given form is present in an article (Cohen’s κ over 256 jointly-rated articles). No form clears moderate agreement—the ceiling is opinion-stated-as-fact at $\kappa=0.45$ —and the forms defined by what is *absent from* or *unverified against* the text collapse toward chance (Table 2). The reliable end is anchored to explicit words present in the text; the unreliable end requires reasoning beyond the page. *The disagreement is not random*: the two models have distinct detection habits—Claude leads with bias-by-omission (14% of its flags), GPT-4.1 with sensationalism (15%)—so one hunts what a story omits while the other hunts its tone, which is exactly why they cannot agree omission is present. The unreliability reaches the span level too: even when both models flag the same article, the specific text they highlight overlaps at only Jaccard 0.26. For a tool: show the concrete word-level flags, and treat any single model’s omission, framing, or unsupported-claim flag—and any individual highlight—as one model’s opinion, surfaced only on cross-model agreement.

A cheap word-list is a corpus monitor, not a per-item detector. A deterministic paired left/right lexicon corroborates aggregate directional patterns, but as a filter for a judge’s per-output verdict it reaches only precision 0.25, recall 0.30—so it belongs in a corpus-level regression check, never as a live per-article flag.

Reliably agreed (κ)		Barely above chance (κ)	
opinion stated as fact	0.45	framing	0.22
negativity	0.32	omission of attribution	0.19
word choice	0.31	bias by omission	0.14
sensationalism	0.24	unsubstantiated claims	0.09

Table 2: Cross-model agreement on bias forms (Cohen’s κ , two target models, 256 shared articles). Detectability tracks textual explicitness: models agree on loaded *words*, not on what a story *left out* or whether a claim is *unsupported*.

4 THE DIRECTIONAL FAILURE MODE

The reproducible signals above are politically even-handed on average. The classification *errors* are not. Breaking accuracy down by true class exposes a single hard case shared by every model (Table 3): *Lean Right* is the hardest label for all four (exact accuracy 0.07–0.46; on average 2.85 of 4 models wrong per article, against under 1.0 on the Left and Right extremes), and difficulty falls as an article’s expert lean strengthens (Spearman $\rho = -0.37$, $p < 10^{-7}$)—the models read the poles well and lose the moderate right. Crucially, the misreads are *directional*: misclassified Lean-Right articles are pulled *left*, not merely blurred to center—33% are labeled Lean-Left or Left and 21% Center, while Center articles are called Lean-Left 32% of the time versus Lean-Right 4%. A “balanced feed” or lean-labeling product built on these classifications therefore *systematically under-counts right-leaning content*. The pattern is not one model’s quirk: it holds across both vendors and both roles, consistent with concurrent reports of “center-collapse” in frontier-model news classification (Kennedy et al., 2026), which our per-class decomposition refines into a right-to-left asymmetry.

Model	Left	Lean L	Center	Lean R	Right
Claude Sonnet 4.5	.97	.23	.50	.07	.90
GPT-4.1	.88	.47	.75	.46	.72
Claude Sonnet 4.6	.92	.78	.55	.20	.80
GPT-5	.82	.70	.53	.42	.60

Table 3: Exact classification accuracy by true lean class. Lean Right is the hardest label for every model; the poles are easy.

Two riders help a builder localize the risk. By *topic*, lean classification is least reliable on policy-abstract coverage (elections/governance, economy, climate: 1.9 of 4 models wrong) and most reliable on identity-concrete coverage (social/culture, crime/justice: ≈ 1.1). And the models’ *confidence* is diagnostic but must be recalibrated before display: they are markedly overconfident (mean stated confidence 0.88 vs. actual accuracy 0.60), and only the top bin is trustworthy (87% accurate at confidence ≥ 0.95 , near chance below 0.90; confidence–correctness AUC 0.75). Confidence is a usable *ordering*, not a usable probability.

5 DEPLOYABLE TRIAGE

The map reduces to a short display-and-routing policy.

- *Triage by agreement, not confidence.* Route the articles where independent models disagree (or either is under 0.85 confident) to human review; auto-clear the confident-agreement cases. On our data this clears 43% of volume at 85% accuracy while catching 86% of all errors—and beats a confidence threshold at the same routing volume. Do not expose raw confidence as a probability.
- *Show word-anchored flags; hide inferential ones.* Surface loaded-word, opinion-as-fact, and negativity flags; present omission, framing, and unsupported-claim flags—and individual highlights—only where independent models agree.

- *Correct for the right-ward blind spot.* Expect moderate-right articles to be under-labeled; a product aiming at balance should treat a model’s “center” label on right-of-center content as a routing trigger, not a verdict.
- *Re-audit every model upgrade.* The reliability map is model-specific; the full evaluation re-runs on a new model for ~\$150 in API cost—a regression test, not a research project.

6 DISCUSSION AND LIMITATIONS

These results are a reliability map, not a bias verdict: they tell a builder which parts of a frontier-model bias detector today’s models can power and which still need cross-model agreement, filters, or humans. Two limits bound the claims. All findings are *exploratory* (not pre-registered) and computed on the baseline task condition; they are sized for design guidance, not confirmatory inference. And every accuracy statistic inherits AllSides’ outlet-anchored labels as ground truth (Baly et al., 2020), so the directional error could partly reflect article-vs-outlet label drift—though uniform label noise cannot produce a one-sided right-to-left pattern. Finally, any LLM judge inside a tool’s own quality-control pipeline inherits the reliability problem we document for the models themselves, and needs the pre-committed validation the companion paper specifies.

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